

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 1 – INTERACTIVE TREE SIMULATOR

|                         |                                                                                                                                                |                      |                                                |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------------------------------------|
| <b>Course Code:</b>     | CSA0309                                                                                                                                        | <b>Course Title:</b> | Data Structures                                |
| <b>CO Assessed:</b>     | CO3 – Precision (BL3, BL5)                                                                                                                     | <b>Assessment:</b>   | Assessment Tool 1 – Interactive Tree Simulator |
| <b>Weightage:</b>       | 30% of CIA                                                                                                                                     | <b>Total Marks:</b>  | 100                                            |
| <b>CO3 Statement:</b>   | <i>Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, BTrees, Red-Black Trees, and Splay Trees.</i> |                      |                                                |
| <b>Student Name:</b>    | VASANTHAN S                                                                                                                                    | <b>Reg. No.:</b>     | 192524458                                      |
| <b>Section / Batch:</b> | B                                                                                                                                              | <b>Date:</b>         | 27/07/2026                                     |

### Code of Conduct

*I VASANTHAN S (192524458) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

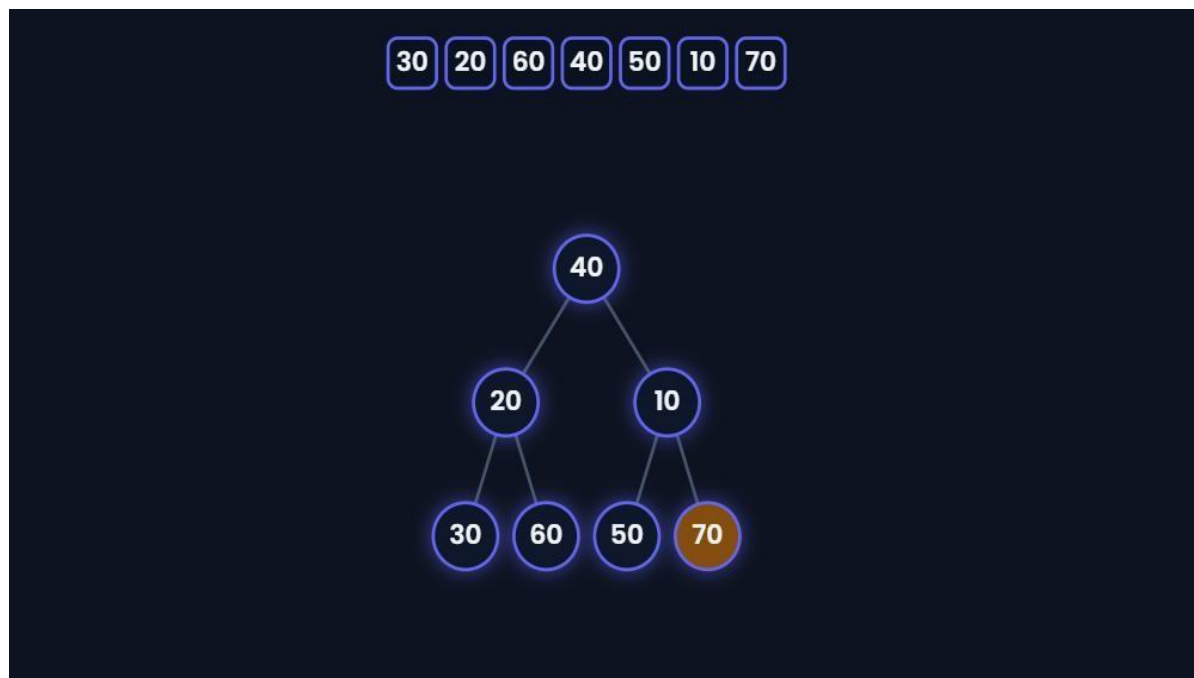
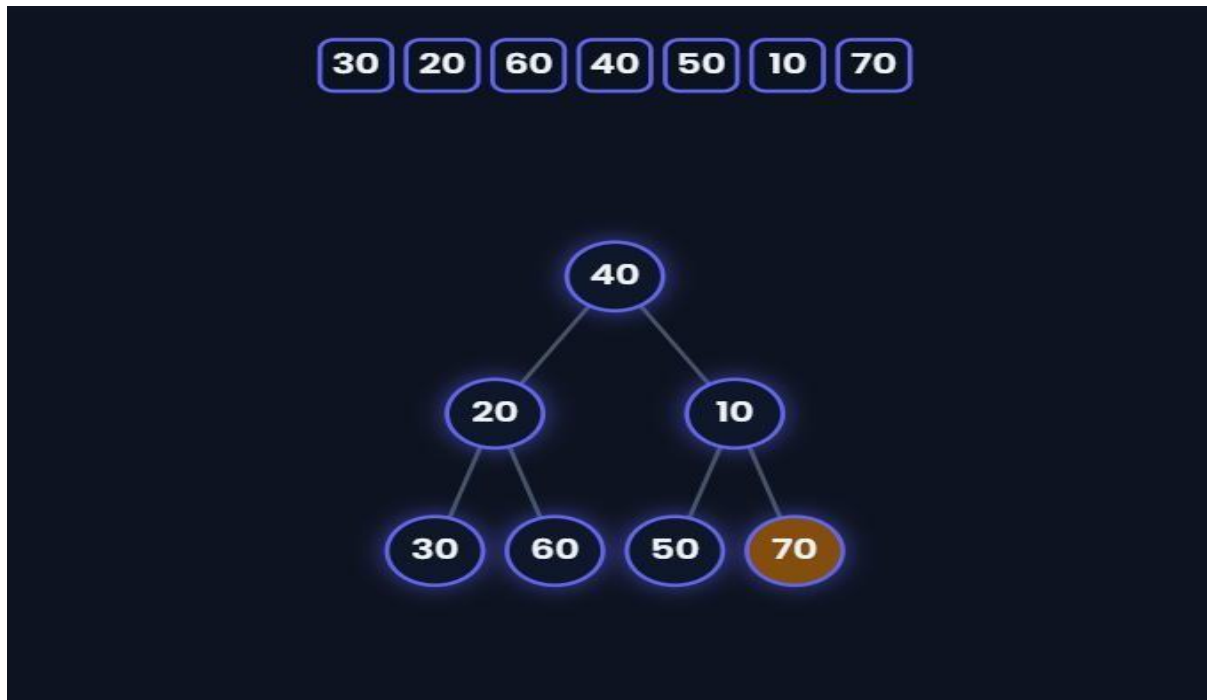
Signature of the Student: VASANTHAN S

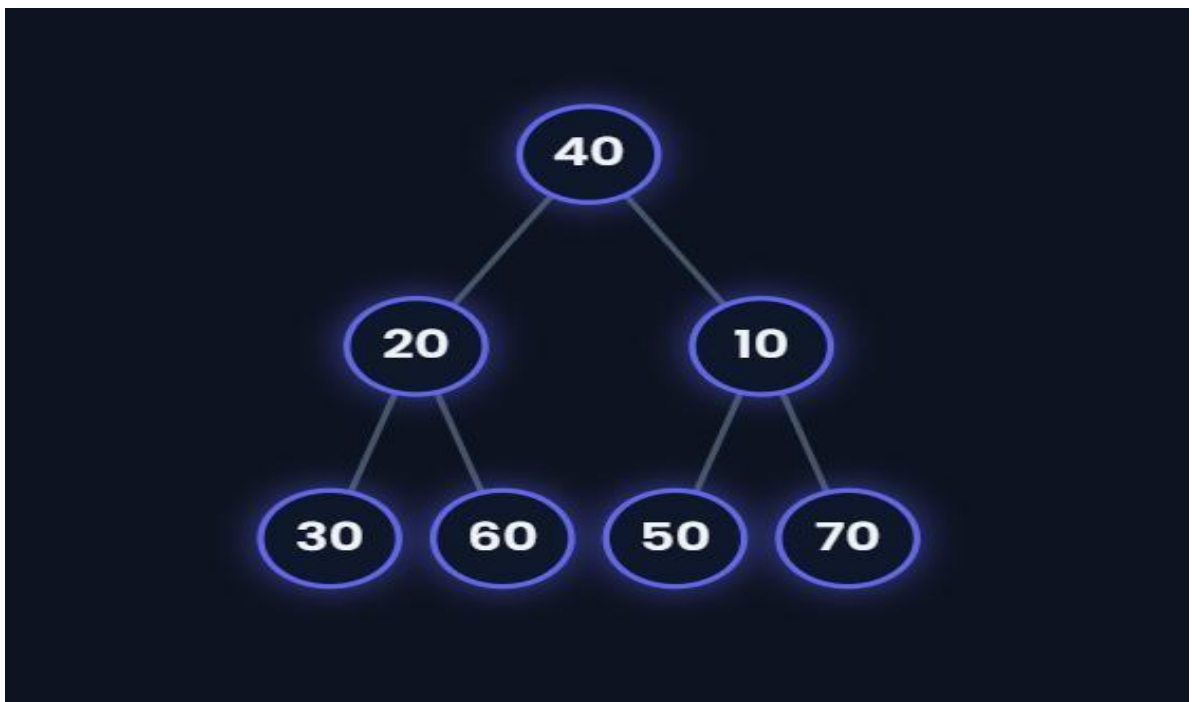
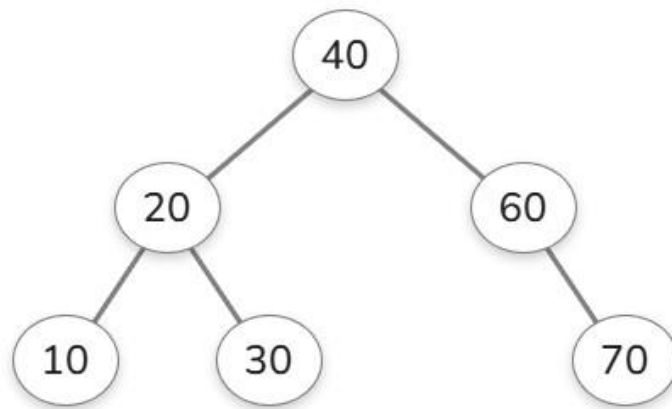
### Instructions

- This assessment tool contains 5 Interactive Tree Simulator questions, Total: 100 Marks.
- Students can use any online/offline Interactive Tree Simulator. Demonstrate insertion, deletion, searching, and traversal operations wherever applicable.
- When a student answer using simulator, in evaluation the answer should have captured screenshots after every major operation.
- Submit the report in PDF format with properly labelled screenshots as proof of completion.

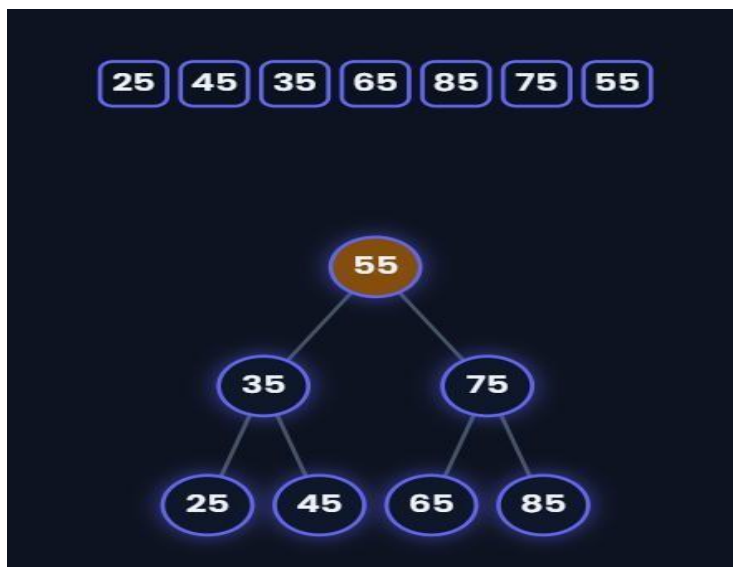
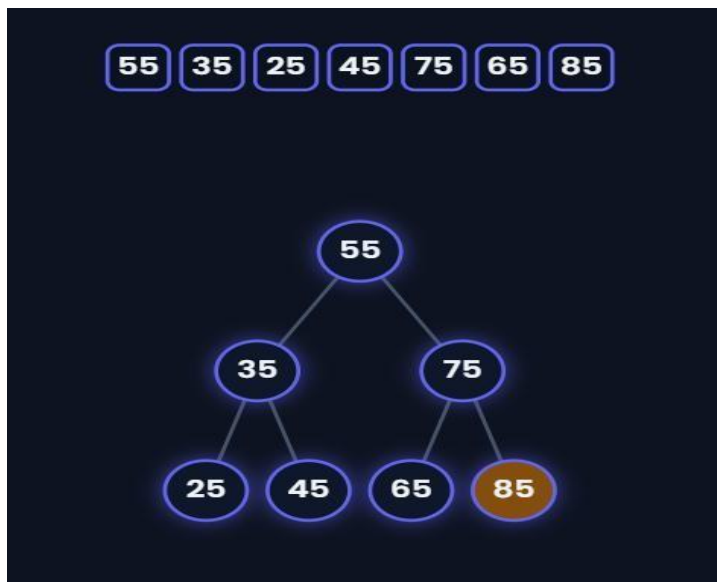
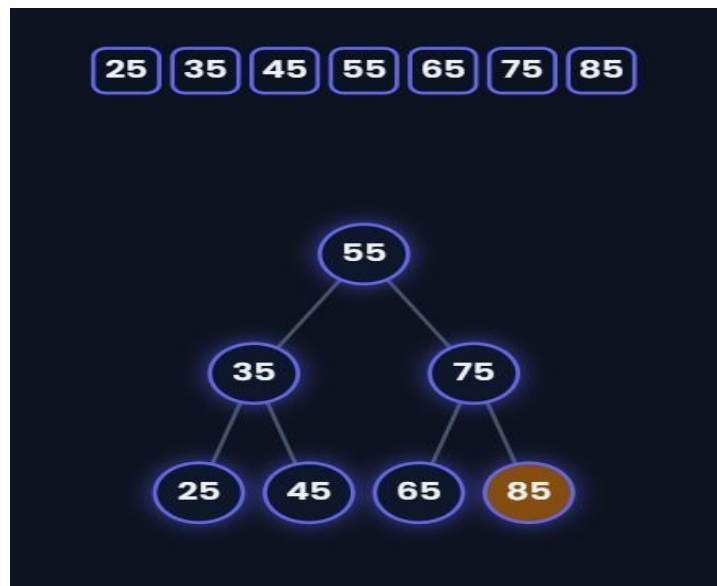
| Section / Topic              | Focus                                                                  | Q Nos | Marks     |
|------------------------------|------------------------------------------------------------------------|-------|-----------|
| Binary Search Tree           | Properties, binary tree and binary search tree, insertion and deletion | 1     | 20        |
| Tree Traversals              | Traversal types, In order, Post order and Pre order                    | 2     | 20        |
| AVL Tree                     | Balancing factor, Rotation, Types of rotation, examples                | 3     | 20        |
| B Tree, 2-3 tree, 2-3-4 tree | Properties, structures and Operations                                  | 4     | 20        |
| Red Black Tree               | Properties, Example and Operations                                     | 5     | 20        |
| GRAND TOTAL:                 |                                                                        |       | 100 Marks |

1. Construct a Binary Search Tree (BST) by inserting the following keys: 40, 20, 10, 30, 60, 50, 70 Perform:

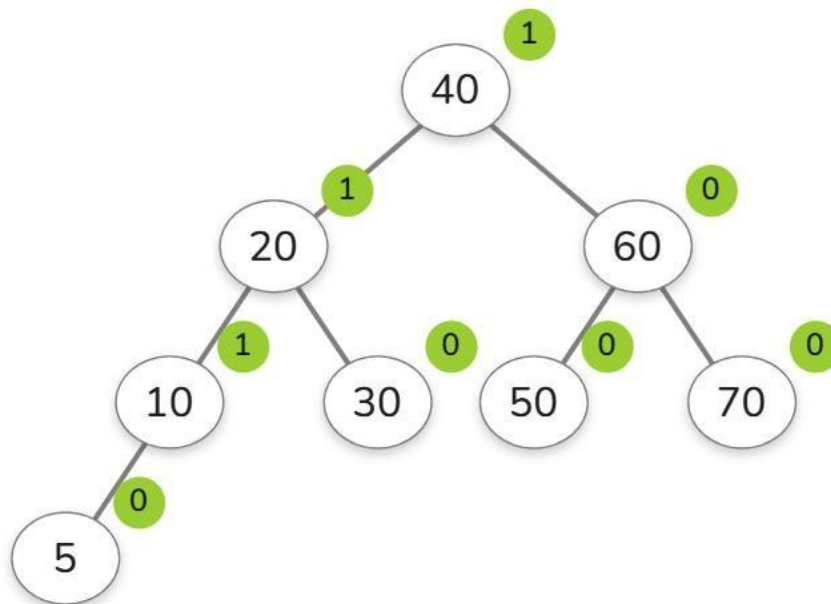




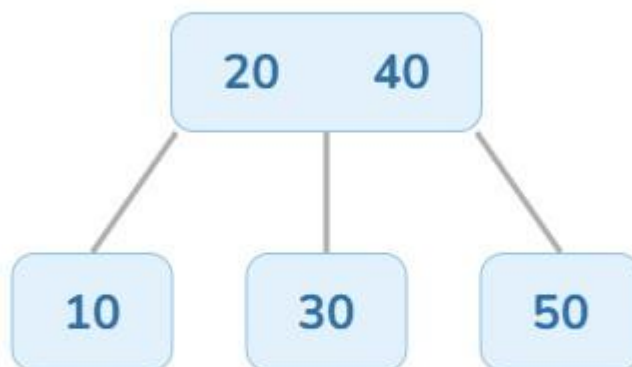
2. Complete Tree Traversals by Inserting 55,35,75,25,45,65,85. Find all the traversals.



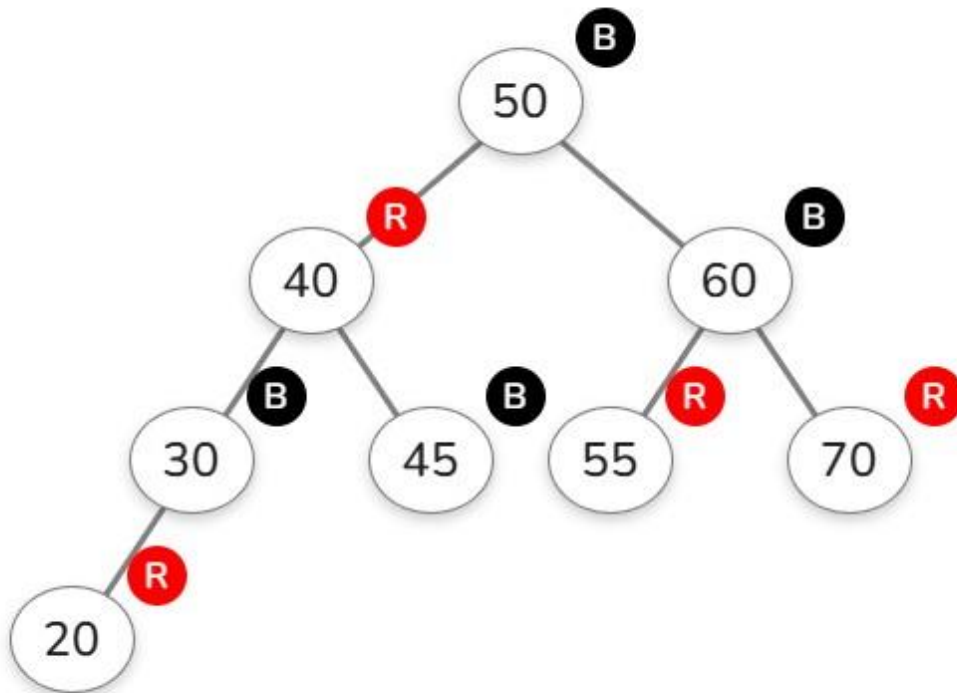
3. Insert 40, 20, 60, 10, 30, 50, 70, 5 into an AVL Tree. Determine the balance factor of every node before and after balancing.



4. Construct a B-Tree of order 3 by inserting 10, 20, 30, 40, 50.



5. **Construct a Red-Black Tree using 50, 40, 60, 30, 45, 55, 70, 20. Delete 30 and analyze how Red-Black properties are restored.**



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## Annexure A – Interactive Tree Simulator

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1. Construct a Binary Search Tree (BST) by inserting the following keys: 40, 20, 10, 30, 60, 50, 70  
Perform:
  - a. Inorder Traversal
  - b. Search for 70
  - c. Delete 50
  - d. Display the final BST.
2. Complete Tree Traversals by Inserting 55,35,75,25,45,65,85. Find all the traversals. a. Inorder  
b. Preorder  
c. Postorder
3. Insert 40, 20, 60, 10, 30, 50, 70, 5 into an AVL Tree. Determine the balance factor of every node before and after balancing.
4. Construct a B-Tree of order 3 by inserting 10, 20, 30, 40, 50.

5. Construct a Red-Black Tree using 50, 40, 60, 30, 45, 55, 70, 20. Delete 30 and analyze how Red-Black properties are restored.

#### Rubrics for Calculation

| Criteria                                | Weightage | Level 4 – Exemplary                                                    | Level 3 – Proficient               | Level 2 – Developing                   | Level 1 – Beginning                |
|-----------------------------------------|-----------|------------------------------------------------------------------------|------------------------------------|----------------------------------------|------------------------------------|
| <b>Correct Tree Construction</b>        | 20        | Tree is constructed correctly with all operations performed accurately | Minor errors in construction       | Some operations incorrect              | Tree construction mostly incorrect |
| <b>Understanding of Tree Operations</b> | 20        | Clearly explains insertion, deletion, search and balancing operations  | Explains most operations correctly | Limited understanding                  | Unable to explain operations       |
| <b>Analysis of Tree Behaviour</b>       | 30        | Correctly analyses balancing, rotations, node split/merge, splaying    | Good analysis with minor omissions | Partial analysis                       | Poor or incorrect analysis         |
| <b>Presentation</b>                     | 20        | Well-organized report with accurate observations and conclusion        | Good report with minor issues      | Basic report with limited observations | Incomplete report                  |
| <b>Accuracy of Responses</b>            | 10        | 90–100% correct answers                                                | 75–89% correct answers             | 50–74% correct answers                 | Below 50% correct answers          |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |